PSET 1: Notation, sets and functions, quadratics, and logarithms

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Translate a statement into notation

A researcher proposes a simple model of protest participation: an individual joins a protest when the benefit they expect from the protest succeeding, weighted by the chance their participation matters, exceeds the personal cost of participating (time, risk of arrest, etc.).

a. Define notation for each quantity in the model (in the style of $R = PB - C$ from the rational voter model in class) and write the condition under which an individual participates.

Ans.:

Notation:

- B = The individual's Expected Benefit from a successful protest.
- P = The Probability that the individual's participation matters.
- C = Personal Cost of participation.
- R = The individual's Net Benefit of participation.

We can define R as follows:

$$\boxed{R = P \times B - C}$$

Condition under which an individual participates:

$$\boxed{R > 0}$$

b. Using your notation, write the formal statement of the following implication: "in a very large protest movement, any one person's participation is unlikely to matter, so individuals should stay home."

Ans.:

Let N denote the size of the protest movement. Based on the given statement, we can infer that as N increases, P reduces (to zero), or as $N \to \infty$, $P \to 0$.

Checking if individuals will stay at home:

$$R = P \times B - C$$
$$= 0 \times B - C$$
$$\boxed{\lim_{N \to \infty} R = -C \text{ or } R < 0}$$

Thus, the individual is better off staying at home as the net benefit of attending the protest is negative.

c. Suppose the government announces harsher penalties for protesters. Which term in your model changes, in which direction, and what does the model predict about participation?

Ans.:

The term affected is C.

Harsher penalties $\implies$ C increases.

Now, looking at the condition for the protest participation (i.e., $R = P \times B - C$), an increase in C will reduce R. So, the model predicts that protest participation will decrease.

Problem 2: Work with sets

Using the sets[1]

$$A = \{2, 3, 7, 9, 12, 13\}$$
$$B = \{x : 6 \leq x \leq 10 \text{ and } x \text{ is an even integer}\} = \{6, 8, 10\}$$
$$C = \{x : 2 < x < 20 \text{ and } x \text{ is prime}\} = \{3, 5, 7, 11, 13, 17, 19\}$$
$$D = \{1, 4, 9, 16, 25, \ldots\}$$
$$E = \{1, 4\}$$

identify the following:

a. $A \cup B$

Ans.: {2, 3, 7, 9, 12, 13, 6, 8, 10}

b. $(A \cup B) \cap C$

Ans.: {3, 7, 13}

c. $C \cap D$

Ans.: ϕ

d. $A \times E$

Ans.: {(2, 1), (3, 1), (7, 1), (9, 1), (12, 1), (13, 1), (2, 4), (3, 4), (7, 4), (9, 4), (12, 4), (13, 4)}

[1] inspired by Grimmer HW1.1

Problem 3: Functions vs. relations

a. True or false—explain why: "All relations are functions, but not all functions are relations."

   Ans.:

   Relations $\implies$ Cartesian Product of Domain and Co-Domain.

   Functions $\implies$ A relation where every element of the Domain is mapped to exactly one element of the Co-Domain.

   So, the statement is **FALSE**, because relations are not restricted on the number of mappings that members of the domain have like functions.

b. For each of the following, state whether y is a function of x (Y/N). If it is a function, state its domain. If it is not, state a restriction that would make it one.

   i. $y = x^2$

      Ans.:

      Yes (any real number has only one square). Domain is $\mathbb{R}$ or $(-\infty, \infty)$.

   ii. $x = y^2$

      Ans.:

      No (since $x = y^2 \implies y = \sqrt{x}$, and square root of 4, for example, can be -2 or +2). Range should be restricted to non-negative real numbers only (i.e., $y \geq 0$).

   iii. $y = x^{1/2}$

      Ans.:

      No (since $y = x^{1/2} \implies y = \sqrt{x}$, and square root of 4, for example, can be -2 or +2). Range should be restricted to non-negative real numbers only (i.e., $y \geq 0$).

c. Is $y = x^2$ invertible on all of $\mathbb{R}$? If not, give a domain on which it is invertible.

   Ans.:

   No, since this fails the horizontal line test OR since multiple values of x can be mapped to the same element of the co-domain (e.g, $(-2)^2 = 4$ and $2^2 = 4$).

   Domain restrictions that make it invertible:

   i. Non-positive real numbers, or $(-\infty, 0]$.
   ii. Non-negative real numbers, or $[0, +\infty)$.

Problem 4: Simplify expressions

Simplify the following expressions as much as possible:[2]

a. $(-x^2y^3)^3$

   Ans.:

$$\boxed{(-x^2y^3)^3 = -x^{2\times3}y^{3\times3} = -x^6y^9}$$

[2] inspired by Gill 1.1

b. $(-2)^{(4-9)}$

Ans.:

$$\boxed{(-2)^{(4-9)} = (-2)^{-5} = -\frac{1}{32}}$$

c. $\left(\dfrac{1}{27b^4}\right)^{1/3}$

Ans.:

$$\begin{aligned}\left(\frac{1}{27b^4}\right)^{1/3} &= \frac{1^{1/3}}{(27b^4)^{1/3}} \\ &= \frac{1}{27^{1/3}b^{4\times 1/3}}\end{aligned}$$

$$\boxed{\left(\frac{1}{27b^4}\right)^{1/3} = \frac{1}{3b^{4/3}} = \frac{1}{3b\sqrt[3]{b}}}$$

d. $\dfrac{13a/7b}{13b/2a}$

Ans.:

$$\boxed{\frac{13a/7b}{13b/2a} = \frac{13a \times 2a}{13b \times 7b} = \frac{2a^2}{7b^2}}1$$

e. $(a+b)^2 + (a-b)^2 + 2(a+b)(a-b) - 3a^2$

Ans.:

$$\begin{aligned}(a+b)^2 + (a-b)^2 + 2(a+b)(a-b) - 3a^2 &= ((a+b)+(a-b))^2 - 3a^2 \\ &= (2a)^2 - 3a^2 \\ &= 4a^2 - 3a^2 \\ &= (4-3)a^2\end{aligned}$$

$$\boxed{(a+b)^2 + (a-b)^2 + 2(a+b)(a-b) - 3a^2 = a^2}$$

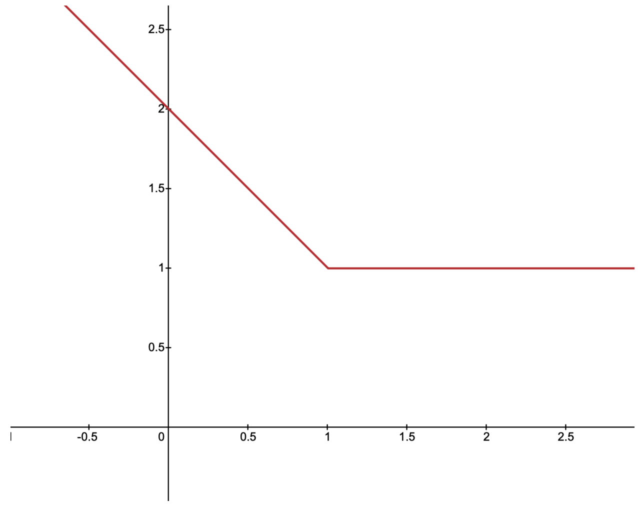
Problem 5: Graph sketching

Let the functions $f(x)$ and $g(x)$ be defined for all $x \in \mathbb{R}$ by[3]

$$f(x) = \begin{cases} |x-2| & \text{if } x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}, \quad g(x) = \begin{cases} x^3 & \text{if } x < 2 \\ 4 & \text{if } x \geq 2 \end{cases}$$

a. Sketch $y = f(x)$

[3] inspired by Pemberton and Rau problem 3-1



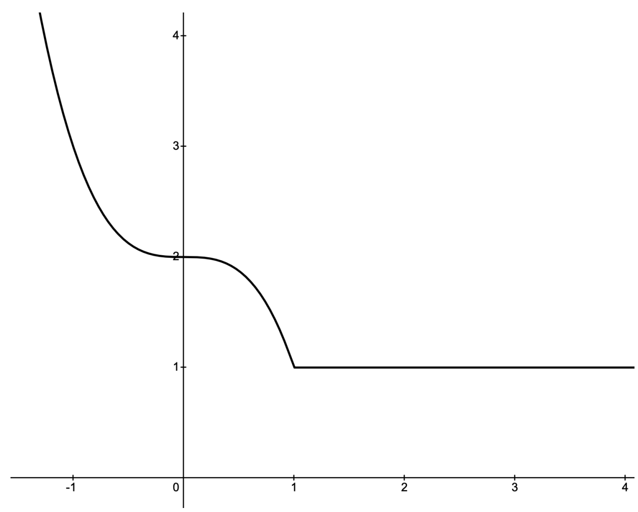
b. Compute $f(g(0))$ and $f(g(3))$

Ans.:

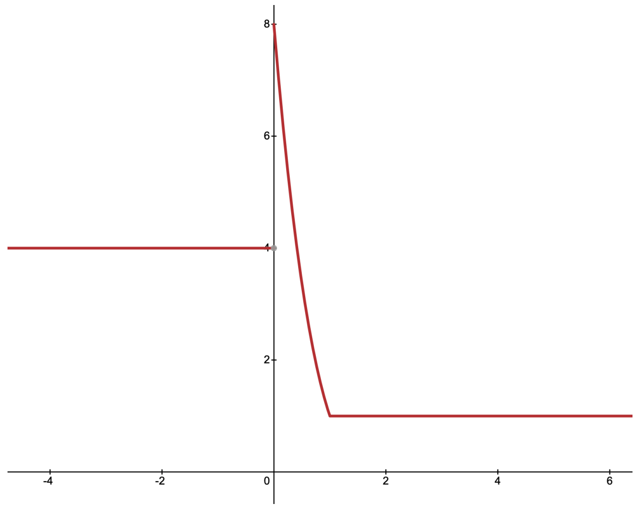
i. $f(g(0)) = f(0^3) = f(0) = |-2| = 2$

ii. $f(g(3)) = f(4) = 1$

c. Sketch $y = f(g(x))$



d. Sketch $y = g(f(x))$



Problem 6: Root finding

Find the roots (solutions) of the following quadratic equations.[4] Recall that for $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

a. $9x^2 - 3x - 12 = 0$

   Ans.:

$$\begin{aligned} 9x^2 - 3x - 12 &= 0 \\ 3x^2 - x - 4 &= 0 \\ 3x^2 + 3x - 4x - 4 &= 0 \\ 3x(x+1) - 4(x+1) &= 0 \\ (3x-4)(x+1) &= 0 \\ \implies x &= \frac{4}{3}, -1 \end{aligned}$$

   are the roots.

b. $x^2 - 2x - 16 = 0$

   Ans.:

$$x^2 - 2x - 16 = 0 \implies a = 1, b = -2, c = -16$$

[4] Gill 1.25

Plugging into the quadratic formula:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-16)}}{2(1)} \\ &= \frac{2 \pm \sqrt{4 + 64}}{2} \\ &= \frac{2 \pm \sqrt{68}}{2} \\ &= \frac{1 \pm \sqrt{17}}{1} \\ &\boxed{x = 1 \pm \sqrt{17}} \end{aligned}$$

are the roots.

c. $6x^2 - 6x - 6 = 0$

Ans.:

$$\begin{aligned} 6x^2 - 6x - 6 = 0 &\implies x^2 - x - 1 = 0 \\ &\implies a = 1, b = -1, c = -1 \end{aligned}$$

Plugging into the quadratic formula:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} \\ &= \frac{1 \pm \sqrt{1 + 4}}{2} \\ &\boxed{x = \frac{1 \pm \sqrt{5}}{2}} \end{aligned}$$

are the roots.

Problem 7: Simplify logarithms

Express each of the following as a single logarithm:[5]

a. $\log(x) - 2\log(y) + \log(z)$

Ans.:

$$\begin{aligned} \log(x) - 2\log(y) + \log(z) &= \log(x) - \log(y^2) + \log(z) \\ &= \log(x) + \log(z) - \log(y^2) \\ &= \log(xz) - \log(y^2) \end{aligned}$$

$$\boxed{\log(x) - 2\log(y) + \log(z) = \log(\frac{xz}{y^2})}$$

[5] Inspired by Grimmer HW1.3. Assume log is natural log unless specified otherwise.

b. $2\log(x) + \log(1)$

Ans.:

$$\boxed{2\log(x) + \log(1) = \log(x^2) + \log(1) = \log(x^2)}$$

c. $\log(2x) - 2$

Ans.:

$$\boxed{\log(2x) - 2 = \log(2x) - \log(e^2) = \log(\frac{2x}{e^2})}$$

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.